Assignment 4
Mod 5 Counter & Band Pass Filter

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1 Mod 5 Counter

1.1 Objective

Our objective is to build a mod 5 counter (should display 0, 1, 2, 3 and 4) using 3 LEDs. The LEDs represent the state (a number from 0-4) in corresponding binary format. For example, the LED states to represent the number 3 would be OFF, ON and ON. The counter will increment after every clock pulse (just after every rising edge). When the counter reaches 4, it loops back to 0.

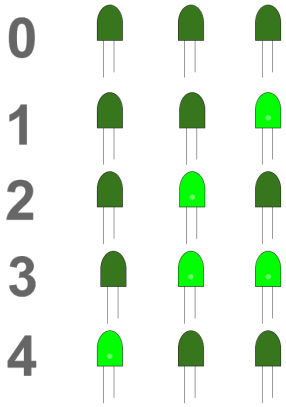


Figure 1.1: Representation of counter states

1.2 Setup

We can achieve this with the help of three JK Flip-Flops (to store the state), and a few AND and NOR gates. We can use a DC Voltage source or an Arduino UNO to supply 5 Volts DC. We can use an AFG to give clock signal(either by giving a square wave of duty cycle 50% and frequency 1Hz, or supplying one square pulse through trigger mode). We can also use an Arduino UNO to supply a clock signal (through any one of the digital pins). In this case, we chose a square wave frequency of 1Hz so that we can see the counter increment properly. If we had applied a higher frequency, we would not be able to see or capture the counter state through the LEDs (the LEDs will turn ON and OFF really quick)

The list of exact components required for the Mod 5 counter are given below:

- 3 - 7476 JK FF ICs

- 1 - 7408 Quad AND Gate IC
- 1 - 7402 Quad NOR Gate IC

1.3 Logic for Counter

We can use a state diagram to represent the states the counter goes through.

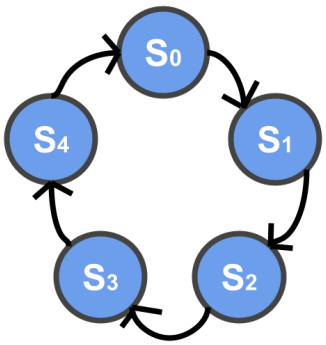


Figure 1.2: State diagram for mod 5 counter

For the 5 states, we would require 3 Flip-Flops (2 Flip-Flops can represent at-most $2^2 = 4$ states and 3 can represent at-most $2^3 = 8$ states. This is of course by assuming that each state is represented as a binary number starting from 0.

The truth table for achieving this is given below.

Q_2	Q_1	Q_0	J_2K_2	J_1K_1	J_0K_0	$Q_{new,2}$	$Q_{new,1}$	$Q_{new,0}$
0	0	0	0 X	0 X	1 X	0	0	1
0	0	1	0 X	1 X	X 1	0	1	0
0	1	0	0 X	X 0	1 X	0	1	1
0	1	1	1 X	X 1	X 1	1	0	0
1	0	0	X 1	0 X	0 X	0	0	0
1	0	1	X 1	0 X	X 1	0	0	0
1	1	0	X 1	X 1	0 X	0	0	0
1	1	1	X 1	X 1	X 1	0	0	0

Table 1.1: Truth table

Here X stands for don't care.

Note that in the above truth table, we made sure that the next Q for undefined states (5,6,7) goes back to 0. This is to ensure that we come back to a defined state even if

we start with an undefined state. If we had left the output as don't cares, then it might take a few iterations (or worse, get stuck) to come out of the set of undefined states.

We can draw K-maps for $J_2, K_2, J_1, K_1, J_0, K_0$, so that we can obtain a simplified expression for the inputs of JK FF :

For J_2:

Q_2 \ Q_1Q_0	00	01	11	10
0	0	0	1	0
1	X	X	X	X

$$J_2 = Q_1Q_0 \qquad (1.1)$$

For K_2:

Q_2 \ Q_1Q_0	00	01	11	10
0	X	X	X	X
1	1	1	1	1

$$K_2 = 1 \qquad (1.2)$$

For J_1:

Q_2 \ Q_1Q_0	00	01	11	10
0	0	1	X	X
1	0	0	X	X

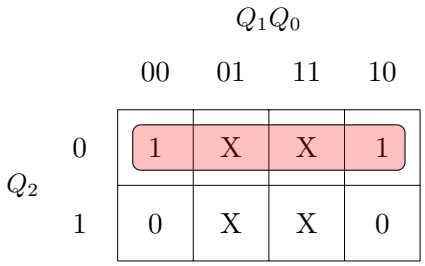
$$J_1 = \overline{Q_2}Q_0 \qquad (1.3)$$

For K_1:

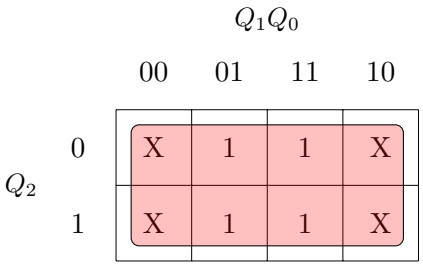
Q_2 \ Q_1Q_0	00	01	11	10
0	X	X	1	0
1	X	X	1	1

$$K_1 = Q_2 + Q_0 \qquad (1.4)$$

For J_0:



For K_0:



$$J_0 = \overline{Q_2} \tag{1.5}$$

$$K_0 = 1 \tag{1.6}$$

1.4 Circuit Diagram

We were initially provided with a 7408 IC (quad AND gate IC). Using the inverted Q outputs that the 7476 JK FF ICs give, we can realize equations (1.1), (1.3) using only AND gates. For equations (1.2), (1.6) and (1.5), direct connections would suffice and no gates are required. For (1.4), an AND gate would not suffice, and hence NOR gates were provided (7402 quad NOR gate IC). The inverted outputs Q_2 and Q_0 were passed through the AND gate, which then goes through the NOR gate with other input being 0(LOW). This is possible because of De Morgan's Law.

$$\bar{Q_2} \text{ AND } \bar{Q_0} = \bar{Q_2}\bar{Q_0} \tag{1.7}$$

$$\bar{Q_2}\bar{Q_0} \text{ NOR } 0 = \overline{\bar{Q_2}\bar{Q_0}} = Q_2 + Q_0 \tag{1.8}$$

Hence we would require a total of 3 AND gates, and 1 NOR gate. Since only quad-gate chips were available to us, we have only used a part of the given chips.

The circuit schematic should look something like this:

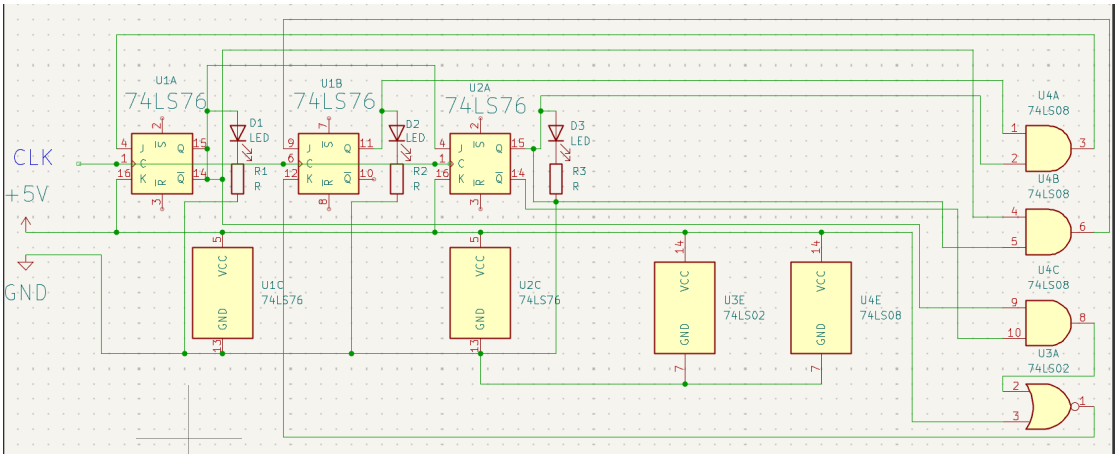


Figure 1.3: Circuit Schematic

It is also important to note that LEDs were not connected directly between Q of JK FF to GND. There was a resistor in between to limit the current flowing, so as to not destroy the LED.

1.5 Observation and Debugging

Initially, the output of the setup was restricted only to 0 and 5. It jumped from 0 to 5 and back to 0. In some cases, the circuit initialized with a state like 2 which then proceeded to 7 then back to 0. So something like this,

$010- > 111- > 000- > 101- > 000- > 101- > 000$ ...

Naturally, our initial guess was that either the logic for the first JK FF was wrong, or we made an error while connecting the output of LED corresponding to MSB (as what should have been 001 became 101). The only expected behavior in this initial setup was the transition from 5 to 0 (101 to 000) and 7 to 0 (111 to 000) just like how we defined it in the truth table. Also initially, what should have been $010- > 011$ became 111, which supports our guess that the first bit was incorrect.

We hence started debugging with an Arduino UNO by connecting probes into random parts of the circuit and looking at the output in the serial monitor. The AND and NOR gates worked as expected, and so did the second and the third Flip-Flop. But there was something off about the first FF (as we initially guessed). The output Q did not match correctly with different J and K. While the logic was right and the value of J_2 and K_2 produced when the state is 000 matches with theoretical expectation, the output Q_2 did not match. So we assumed that the first Flip-Flop itself is not behaving as expected. We immediately checked the clear and preset pins, and they were HIGH as expected (the clear and preset pins are active-LOW). So we assumed that either the wiring for the rest of the first flip-flop isn't proper or the flip-flop itself was faulty. We hence rewired only the first flip-flop IC and also replaced it with a new one. After then fixing a few wiring mismatches, the mod 5 counter started working as expected.

1.6 Conclusion

The LEDs were blinking in this fashion after every 1 second (clock signal was produced by AFG): $000- > 001- > 010- > 011- > 100- > 000$ We did not encounter any undefined state initially in this refined setup. We have hence successfully built a mod 5 counter with a nice representation through LEDs.

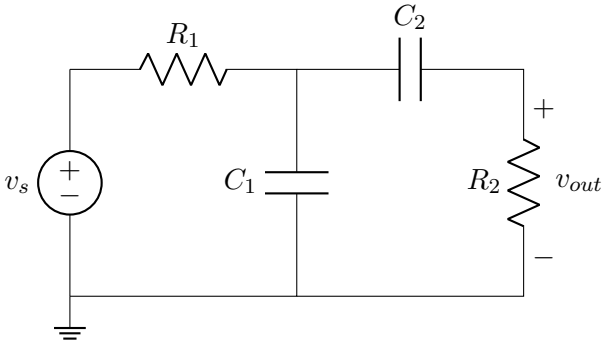
2 Band Pass Filter

2.1 Objective

A band pass filter is a filter that attenuates all frequencies other than a specific band. We build a band pass filter by cascading a low pass filter with a high pass filter. The frequency response should be centered at $15k$Hz, and we must try to extract the $15k$Hz harmonic from a $3k$Hz square wave.

2.2 Calculations and setup

The general circuit diagram for a band pass filter that works by cascading a low pass filter followed by a high pass filter is as follows:



$$\frac{v_{out}}{v_s} = \frac{1}{1 + j\omega R_1 C_1} \cdot \frac{j\omega R_2 C_2}{1 + j\omega R_2 C_2} \tag{2.1}$$

If our goal is to maximize magnitude output to input ratio at a particular frequency (in this case $15k$Hz) we differentiate the above expression w.r.t ω and set it to 0, to find maxima.

$$\left|\frac{v_{out}}{v_s}\right| = \frac{1}{\sqrt{1 + w^2 R_1^2 C_1^2}} \cdot \frac{w R_2 C_2}{\sqrt{1 + w^2 R_2 C_2}} \tag{2.2}$$

$$\frac{d}{d\omega}\left(\frac{1}{\sqrt{1 + w^2 R_1^2 C_1^2}} \cdot \frac{w R_2 C_2}{\sqrt{1 + w^2 R_2 C_2}}\right) = 0 \tag{2.3}$$

$$\omega^2 - R_1 C_1 R_2 C_2 = 0 \tag{2.4}$$

We hence end up with,

$$\omega_0 = \frac{1}{\sqrt{R_1 C_1 R_2 C_2}} \tag{2.5}$$

Where ω_0 is the angular frequency corresponding to where the frequency response is centered. We can interpret this central angular frequency as the geometric ratio of the cutoff frequencies of low-pass and high-pass filters.

Now we must choose R_1, C_1, R_2 and C_2 such that $\frac{\omega_0}{2\pi} = 15k\text{Hz}$ will be satisfied.

We must also take care of the distance of the cutoff frequencies of the low-pass and high-pass component of the band pass filter.

In this case, our goal is to design a narrow band pass filter, so we must choose the cutoff frequencies of the corresponding low and high pass filter circuits such that they are as close to each other as possible. We hence took the cutoff frequencies to be the same, i.e $R_1 C_1 = R_2 C_2$

So if we set $R_1 C_1 = R_2 C_2$, we get

$$RC = \frac{1}{2\pi \times 15 \times 10^3} \tag{2.6}$$

$$RC = 1.06 \times 10^{-5} \tag{2.7}$$

So our goal is to find corresponding values of R and C based on the resistors and capacitors available to us. We initially considered using a $1\mu F$ capacitor with a 10Ω resistor. But we then realized that the AFG itself has an impedance of 50Ω. This would mess up the true value of RC, and hence the frequency response would not be centered at $15k$Hz.

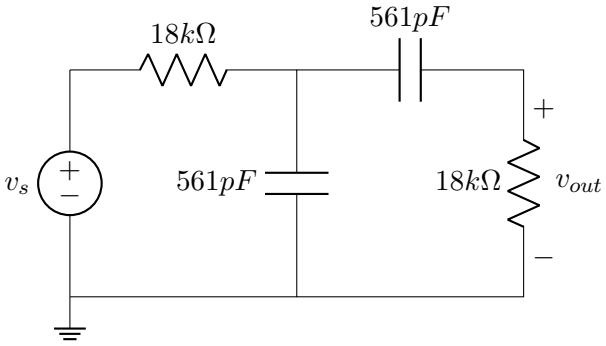
So we tried to maximize R so that the 50Ω impedance would have a negligible effect on overall R. Since there weren't any capacitors with capacitances in the order of nF, we used the highest capacitance in the pF range, which was $561pF$. We then calculated R so that $RC \approx 10^{-5}$. R came out to be $18.8k\Omega$, which we were lucky to have. We decided to use two $18k\Omega$ resistors and two $561pF$ capacitors in our circuit.

Element	Value
R_1	$18k\Omega$
C_1	$561pF$
R_2	$18k\Omega$
C_2	$561pF$

Table 2.1: Resistors and Capacitors used

2.3 Circuit Diagram

The final circuit diagram ended up looking something like this:



Terminals of AFG were used as the voltage source v_s. Probes of the DSO were connected between the terminals of the resistor corresponding to v_{out}.

2.4 Output and Simulations

Figure 2.1: Output Waveform

The output waveform unfortunately did not look like a clean sine wave. We tried checking if the frequency response was centered at $15k$Hz, and it was indeed matching with the theoretical expectation.

We simulated the output waveform, and it DID match with the experimental response.

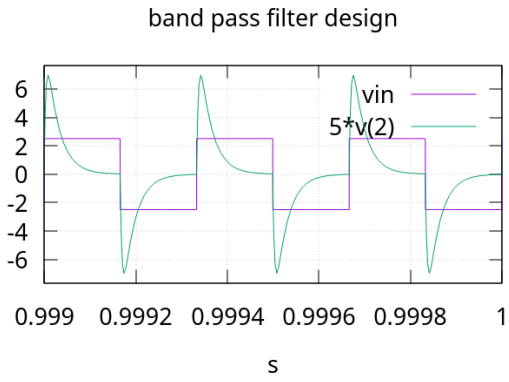


Figure 2.2: Simulated Time-Domain Response to 3kHz square wave

We also simulated the frequency response, and it was matching with the experimental results. It was centered at around $15 - 17kHz$ (we used the exact resistances that we used in the experiment to simulate).

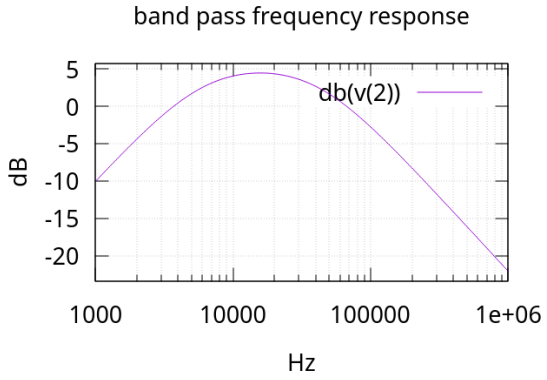


Figure 2.3: Simulated Frequency-Domain Response

2.5 Observation

In a square wave, the fourier series coefficients are given as follows

$$C_n = \begin{cases} \frac{2}{nj\pi} & \text{if n is odd} \\ 0 & \text{if n is even} \end{cases} \tag{2.8}$$

Note that in the equation above, n can be negative.

So for every odd harmonic, the coefficient decays as $\frac{1}{n}$. So the $15k$Hz or the 5^{th} harmonic has only 1/5 weightage of the first harmonic ($3k$Hz). The band pass filter is not able to attenuate the $3k$Hz frequency enough to remove its influence. Hence the

output waveform also looks like it has a frequency of $3kHz$, even though it has a good $15k$Hz component.

2.6 Conclusion

Band pass filters built by using a low pass filter followed by a high pass filter are great for attenuating frequencies other than wide bands. But in this case, where we are expected to extract only the $15kHz$ sine wave, the band pass filter failed. However we were able to get the frequency response to be centered at $15kHz$.

All the simulations are available here:

https://github.com/Vivek-Kumar-1907/ee1200/tree/main/assignment4/simulations/